a02_4

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[1]:
```
import pandas as pd
import numpy as np
import torch
import matplotlib.pyplot as plt
from itertools import product

from a02_functions import SimpleCNN, train_model
from a02_helper import get_raw_data, count_model_params

%load_ext autoreload
%autoreload 2
```

1 Task 4: Exploration

[2]:
```
data = get_raw_data()

def run_sweep(data, model_fn, configs: list[dict], sweep_name: str = "") ->␣
↪dict[str, dict]:
    all_results = {}
    for cfg in configs:
        label = ", ".join(f"{k}={v}" for k, v in cfg.items())
        print(f"\n--- [{sweep_name}] Config: {label} ---")
        model = model_fn()
        results = train_model(data, model, **cfg)
        results["num_params"] = count_model_params(model)
        all_results[label] = results
    return all_results


def plot_sweep(all_results: dict, metric: str = "val_acc", eval_every: int =␣
↪10, title_prefix: str = ""):
    EPOCH_METRICS = {"train_acc", "val_acc", "train_losses", "val_losses"}
    if metric not in EPOCH_METRICS:
```

```
        raise ValueError(f"'{metric}' is a scalar. Use one of:␣
↪{sorted(EPOCH_METRICS)}")

    loss_key = "val_losses" if "val" in metric else "train_losses"
    fig, axes = plt.subplots(1, 2, figsize=(12, 4))
    for label, res in all_results.items():
        epochs = [i * eval_every for i in range(len(res[metric]))]
        axes[0].plot(epochs, res[loss_key], label=label)
        axes[1].plot(epochs, res[metric], label=label)

    axes[0].set(title=f"{title_prefix} Loss", xlabel="Epoch", ylabel="Loss")
    axes[1].set(title=f"{title_prefix} Accuracy", xlabel="Epoch",␣
↪ylabel="Accuracy (%)")
    for ax in axes:
        ax.legend(fontsize=8)
        ax.grid(alpha=0.3)
    plt.tight_layout()
    plt.show()


def plot_final_comparison(all_results: dict, metric: str = "test_acc",␣
↪title_prefix: str = ""):
    labels = list(all_results.keys())
    values = [res[metric][0] for res in all_results.values()]

    fig, ax = plt.subplots(figsize=(8, 4))
    ax.bar(labels, values)
    ax.set(title=f"{title_prefix} - Final {metric} by config", ylabel=metric)
    ax.tick_params(axis="x", rotation=30)
    plt.tight_layout()
    plt.show()


def build_summary_table(sweep_results: dict[str, dict]) -> pd.DataFrame:
    """
    sweep_results: { sweep_name: { config_label: result_dict } }
    Returns a DataFrame with one row per config, columns:
      sweep | config | test_acc | val_acc (final) | train_acc (final) |␣
↪val_loss (final) | train_loss (final)
    """
    rows = []
    for sweep_name, all_results in sweep_results.items():
        for label, res in all_results.items():
            rows.append({
                "sweep":            sweep_name,
                "config":           label,
                "test_acc":         res["test_acc"][0],
```

```
                "val_acc_final":    res["val_acc"][-1],
                "train_acc_final":  res["train_acc"][-1],
                "val_loss_final":   res["val_losses"][-1],
                "train_loss_final": res["train_losses"][-1],
                "num_params":      res["num_params"],
            })
    df = pd.DataFrame(rows)
    df = df.sort_values(["sweep", "test_acc"], ascending=[True, False]).
 ↪reset_index(drop=True)
    return df
```

1.1 Run all sweeps

[3]:
```
sweep_results = {}

# --- Learning rate sweep ---
lrs = np.logspace(-5, 1, 5)
configs = [{"lr": float(lr)} for lr in lrs]
sweep_results["lr"] = run_sweep(data, lambda: SimpleCNN(), configs,␣
 ↪sweep_name="lr")
plot_sweep(sweep_results["lr"], metric="val_acc", title_prefix="LR Sweep")
plot_final_comparison(sweep_results["lr"], metric="test_acc", title_prefix="LR␣
 ↪Sweep")

# --- Batch size sweep ---
batch_sizes = [2**i for i in range(4, 9)]
configs = [{"batch_size": int(bs)} for bs in batch_sizes]
sweep_results["batch_size"] = run_sweep(data, lambda: SimpleCNN(), configs,␣
 ↪sweep_name="batch_size")
plot_sweep(sweep_results["batch_size"], metric="val_acc", title_prefix="Batch␣
 ↪Size Sweep")
plot_final_comparison(sweep_results["batch_size"], metric="test_acc",␣
 ↪title_prefix="Batch Size Sweep")

# --- Epochs sweep ---
epoch_list = [10, 50, 100, 200, 500]
configs = [{"epochs": int(e)} for e in epoch_list]
sweep_results["epochs"] = run_sweep(data, lambda: SimpleCNN(), configs,␣
 ↪sweep_name="epochs")
plot_sweep(sweep_results["epochs"], metric="val_acc", title_prefix="Epochs␣
 ↪Sweep")
plot_final_comparison(sweep_results["epochs"], metric="test_acc",␣
 ↪title_prefix="Epochs Sweep")
```

```
--- [lr] Config: lr=9.999999999999999e-06 ---
```

```
epoch 0, dt 0.91s, train_loss 2.300e+00, val_loss 2.309e+00, train_acc 4.3,
val_acc 4.6
epoch 10, dt 1.39s, train_loss 2.282e+00, val_loss 2.303e+00, train_acc 6.0,
val_acc 7.0
epoch 20, dt 1.45s, train_loss 2.291e+00, val_loss 2.297e+00, train_acc 9.5,
val_acc 10.1
epoch 30, dt 1.40s, train_loss 2.279e+00, val_loss 2.287e+00, train_acc 16.9,
val_acc 17.4
epoch 40, dt 1.40s, train_loss 2.298e+00, val_loss 2.268e+00, train_acc 17.2,
val_acc 18.0
epoch 50, dt 1.37s, train_loss 2.234e+00, val_loss 2.236e+00, train_acc 19.2,
val_acc 20.5
epoch 60, dt 1.41s, train_loss 2.204e+00, val_loss 2.187e+00, train_acc 19.8,
val_acc 22.0
epoch 70, dt 1.44s, train_loss 2.144e+00, val_loss 2.121e+00, train_acc 20.9,
val_acc 21.2
epoch 80, dt 1.43s, train_loss 2.112e+00, val_loss 2.053e+00, train_acc 20.9,
val_acc 21.1
epoch 90, dt 1.47s, train_loss 2.073e+00, val_loss 1.991e+00, train_acc 21.3,
val_acc 21.1
Final result: train_loss 1.876e+00, val_loss 1.991e+00, test_loss 1.933e+00,
train_acc 21.3, val_acc 21.1, test_acc 21.4

--- [lr] Config: lr=0.00031622776601683794 ---
epoch 0, dt 0.14s, train_loss 2.287e+00, val_loss 2.294e+00, train_acc 17.0,
val_acc 16.5
epoch 10, dt 1.45s, train_loss 1.395e+00, val_loss 1.454e+00, train_acc 44.8,
val_acc 40.9
epoch 20, dt 1.65s, train_loss 1.235e+00, val_loss 1.333e+00, train_acc 50.8,
val_acc 46.9
epoch 30, dt 1.49s, train_loss 1.059e+00, val_loss 1.263e+00, train_acc 54.2,
val_acc 50.4
epoch 40, dt 1.56s, train_loss 1.214e+00, val_loss 1.230e+00, train_acc 55.8,
val_acc 49.5
epoch 50, dt 1.65s, train_loss 1.158e+00, val_loss 1.182e+00, train_acc 56.6,
val_acc 51.9
epoch 60, dt 1.87s, train_loss 8.941e-01, val_loss 1.139e+00, train_acc 60.1,
val_acc 53.0
epoch 70, dt 1.77s, train_loss 1.045e+00, val_loss 1.117e+00, train_acc 61.3,
val_acc 53.9
epoch 80, dt 1.99s, train_loss 8.987e-01, val_loss 1.084e+00, train_acc 62.0,
val_acc 53.1
epoch 90, dt 1.89s, train_loss 8.579e-01, val_loss 1.062e+00, train_acc 65.0,
val_acc 55.4
Final result: train_loss 7.112e-01, val_loss 1.062e+00, test_loss 9.693e-01,
train_acc 65.0, val_acc 55.4, test_acc 58.4

--- [lr] Config: lr=0.01 ---
```

```
epoch 0, dt 0.15s, train_loss 1.436e+00, val_loss 1.499e+00, train_acc 37.2,
val_acc 34.5
epoch 10, dt 1.46s, train_loss 8.784e-01, val_loss 5.991e-01, train_acc 84.0,
val_acc 78.2
epoch 20, dt 1.72s, train_loss 1.244e-01, val_loss 5.815e-01, train_acc 93.0,
val_acc 84.0
epoch 30, dt 1.50s, train_loss 2.646e-01, val_loss 5.403e-01, train_acc 95.5,
val_acc 85.5
epoch 40, dt 1.59s, train_loss 5.804e-02, val_loss 6.085e-01, train_acc 98.9,
val_acc 88.2
epoch 50, dt 1.62s, train_loss 5.003e-02, val_loss 5.364e-01, train_acc 98.9,
val_acc 87.2
epoch 60, dt 2.78s, train_loss 2.830e-02, val_loss 6.000e-01, train_acc 98.2,
val_acc 88.1
epoch 70, dt 2.30s, train_loss 9.289e-02, val_loss 7.032e-01, train_acc 97.7,
val_acc 88.0
epoch 80, dt 1.83s, train_loss 1.467e-02, val_loss 7.755e-01, train_acc 98.6,
val_acc 88.6
epoch 90, dt 1.74s, train_loss 6.250e-02, val_loss 9.060e-01, train_acc 97.6,
val_acc 86.6
Final result: train_loss 9.663e-02, val_loss 9.060e-01, test_loss 7.185e-01,
train_acc 97.6, val_acc 86.6, test_acc 88.8

--- [lr] Config: lr=0.31622776601683794 ---
epoch 0, dt 0.20s, train_loss 2.336e+00, val_loss 2.330e+00, train_acc 10.1,
val_acc 10.0
epoch 10, dt 1.73s, train_loss 2.325e+00, val_loss 2.321e+00, train_acc 10.1,
val_acc 10.0
epoch 20, dt 1.59s, train_loss 2.322e+00, val_loss 2.335e+00, train_acc 9.8,
val_acc 9.9
epoch 30, dt 1.46s, train_loss 2.319e+00, val_loss 2.329e+00, train_acc 10.1,
val_acc 10.0
epoch 40, dt 1.69s, train_loss 2.302e+00, val_loss 2.318e+00, train_acc 10.0,
val_acc 10.0
epoch 50, dt 1.79s, train_loss 2.325e+00, val_loss 2.312e+00, train_acc 10.1,
val_acc 10.0
epoch 60, dt 1.79s, train_loss 2.308e+00, val_loss 2.315e+00, train_acc 9.9,
val_acc 10.0
epoch 70, dt 1.51s, train_loss 2.327e+00, val_loss 2.341e+00, train_acc 10.1,
val_acc 10.0
epoch 80, dt 1.56s, train_loss 2.300e+00, val_loss 2.311e+00, train_acc 10.3,
val_acc 10.2
epoch 90, dt 1.50s, train_loss 2.330e+00, val_loss 2.319e+00, train_acc 9.9,
val_acc 10.0
Final result: train_loss 2.338e+00, val_loss 2.319e+00, test_loss 2.320e+00,
train_acc 9.9, val_acc 10.0, test_acc 10.6

--- [lr] Config: lr=10.0 ---
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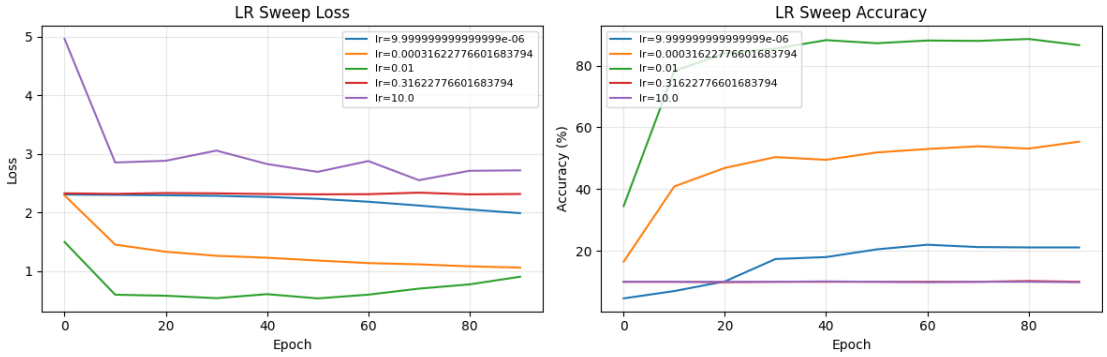
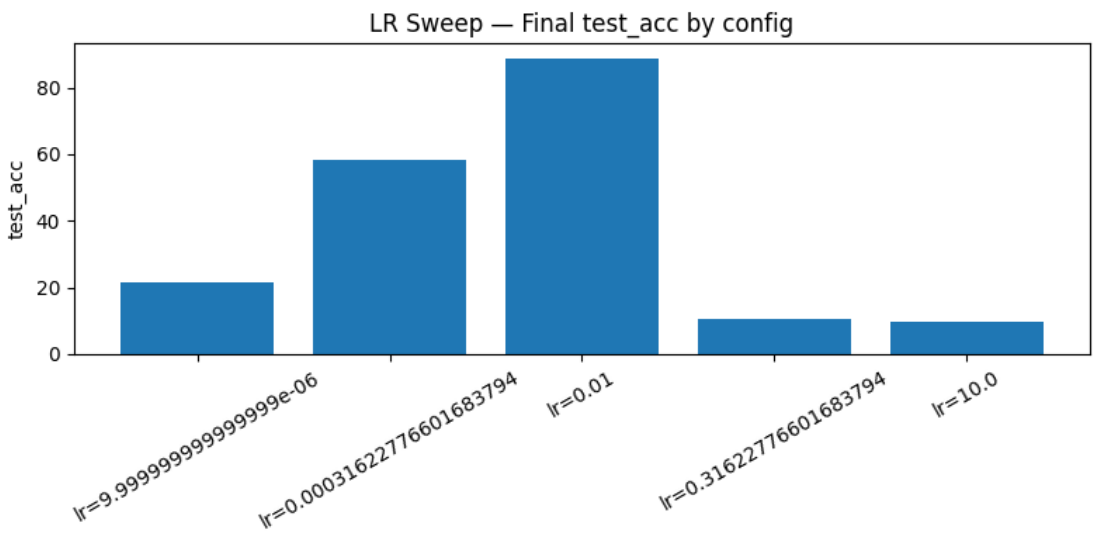
```
epoch 0, dt 0.17s, train_loss 5.202e+00, val_loss 4.964e+00, train_acc 10.1,
val_acc 10.0
epoch 10, dt 1.68s, train_loss 2.815e+00, val_loss 2.856e+00, train_acc 10.1,
val_acc 10.0
epoch 20, dt 1.56s, train_loss 2.989e+00, val_loss 2.885e+00, train_acc 10.1,
val_acc 10.0
epoch 30, dt 1.55s, train_loss 2.832e+00, val_loss 3.059e+00, train_acc 10.0,
val_acc 10.0
epoch 40, dt 1.82s, train_loss 2.563e+00, val_loss 2.830e+00, train_acc 10.1,
val_acc 10.1
epoch 50, dt 1.55s, train_loss 3.038e+00, val_loss 2.696e+00, train_acc 9.9,
val_acc 10.0
epoch 60, dt 1.61s, train_loss 3.021e+00, val_loss 2.881e+00, train_acc 9.9,
val_acc 9.9
epoch 70, dt 1.65s, train_loss 2.945e+00, val_loss 2.554e+00, train_acc 10.0,
val_acc 10.0
epoch 80, dt 1.85s, train_loss 2.508e+00, val_loss 2.714e+00, train_acc 10.1,
val_acc 10.0
epoch 90, dt 1.89s, train_loss 2.566e+00, val_loss 2.722e+00, train_acc 9.8,
val_acc 9.9
Final result: train_loss 2.589e+00, val_loss 2.722e+00, test_loss 3.183e+00,
train_acc 9.8, val_acc 9.9, test_acc 9.8
```






```
--- [batch_size] Config: batch_size=16 ---
epoch 0, dt 0.57s, train_loss 1.834e+00, val_loss 1.384e+00, train_acc 43.0,
val_acc 40.6
epoch 10, dt 6.16s, train_loss 6.666e-01, val_loss 6.221e-01, train_acc 85.7,
val_acc 81.6
epoch 20, dt 5.41s, train_loss 2.573e-01, val_loss 6.348e-01, train_acc 89.7,
val_acc 83.1
epoch 30, dt 5.91s, train_loss 1.628e-02, val_loss 4.910e-01, train_acc 94.4,
val_acc 86.4
epoch 40, dt 6.37s, train_loss 1.635e-01, val_loss 5.607e-01, train_acc 94.6,
val_acc 88.9
epoch 50, dt 6.25s, train_loss 2.963e-01, val_loss 5.270e-01, train_acc 93.7,
val_acc 85.6
epoch 60, dt 5.72s, train_loss 8.754e-02, val_loss 6.250e-01, train_acc 92.5,
val_acc 85.2
epoch 70, dt 5.92s, train_loss 7.667e-03, val_loss 5.509e-01, train_acc 94.2,
val_acc 87.4
epoch 80, dt 6.38s, train_loss 2.580e-02, val_loss 5.654e-01, train_acc 96.3,
val_acc 87.5
epoch 90, dt 6.73s, train_loss 1.681e-05, val_loss 5.899e-01, train_acc 98.0,
val_acc 88.9
Final result: train_loss 1.388e-01, val_loss 5.899e-01, test_loss 9.278e-01,
train_acc 98.0, val_acc 88.9, test_acc 83.6

--- [batch_size] Config: batch_size=32 ---
epoch 0, dt 0.28s, train_loss 1.304e+00, val_loss 1.500e+00, train_acc 40.4,
val_acc 38.4
epoch 10, dt 3.00s, train_loss 8.707e-01, val_loss 9.053e-01, train_acc 69.4,
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```
val_acc 61.2
epoch 20, dt 3.29s, train_loss 5.610e-01, val_loss 8.140e-01, train_acc 80.8,
val_acc 71.2
epoch 30, dt 3.52s, train_loss 6.569e-01, val_loss 7.058e-01, train_acc 88.8,
val_acc 78.9
epoch 40, dt 3.43s, train_loss 1.017e-01, val_loss 8.488e-01, train_acc 88.5,
val_acc 78.8
epoch 50, dt 3.58s, train_loss 1.783e-01, val_loss 7.311e-01, train_acc 94.4,
val_acc 82.0
epoch 60, dt 3.25s, train_loss 1.630e-01, val_loss 9.001e-01, train_acc 90.2,
val_acc 80.2
epoch 70, dt 3.29s, train_loss 3.026e-01, val_loss 8.165e-01, train_acc 92.7,
val_acc 82.5
epoch 80, dt 2.96s, train_loss 1.865e-01, val_loss 8.672e-01, train_acc 94.5,
val_acc 82.6
epoch 90, dt 2.75s, train_loss 1.993e-01, val_loss 7.484e-01, train_acc 95.7,
val_acc 84.4
Final result: train_loss 5.148e-01, val_loss 7.484e-01, test_loss 7.622e-01,
train_acc 95.7, val_acc 84.4, test_acc 84.5

--- [batch_size] Config: batch_size=64 ---
epoch 0, dt 0.18s, train_loss 1.465e+00, val_loss 1.534e+00, train_acc 36.7,
val_acc 35.5
epoch 10, dt 1.58s, train_loss 5.041e-01, val_loss 6.546e-01, train_acc 81.6,
val_acc 75.6
epoch 20, dt 1.62s, train_loss 3.108e-01, val_loss 4.943e-01, train_acc 94.0,
val_acc 86.0
epoch 30, dt 1.93s, train_loss 8.960e-02, val_loss 5.604e-01, train_acc 95.8,
val_acc 85.4
epoch 40, dt 1.93s, train_loss 9.490e-02, val_loss 6.030e-01, train_acc 97.5,
val_acc 86.1
epoch 50, dt 1.63s, train_loss 1.025e-01, val_loss 5.183e-01, train_acc 98.1,
val_acc 89.0
epoch 60, dt 2.07s, train_loss 1.928e-01, val_loss 5.888e-01, train_acc 97.8,
val_acc 87.9
epoch 70, dt 1.88s, train_loss 6.065e-02, val_loss 9.072e-01, train_acc 96.9,
val_acc 86.8
epoch 80, dt 2.55s, train_loss 3.171e-03, val_loss 8.579e-01, train_acc 99.7,
val_acc 88.4
epoch 90, dt 1.73s, train_loss 3.058e-02, val_loss 9.696e-01, train_acc 97.0,
val_acc 86.9
Final result: train_loss 1.914e-02, val_loss 9.696e-01, test_loss 8.189e-01,
train_acc 97.0, val_acc 86.9, test_acc 86.5

--- [batch_size] Config: batch_size=128 ---
epoch 0, dt 0.38s, train_loss 1.638e+00, val_loss 1.724e+00, train_acc 30.3,
val_acc 27.4
epoch 10, dt 1.28s, train_loss 7.881e-01, val_loss 8.809e-01, train_acc 68.1,
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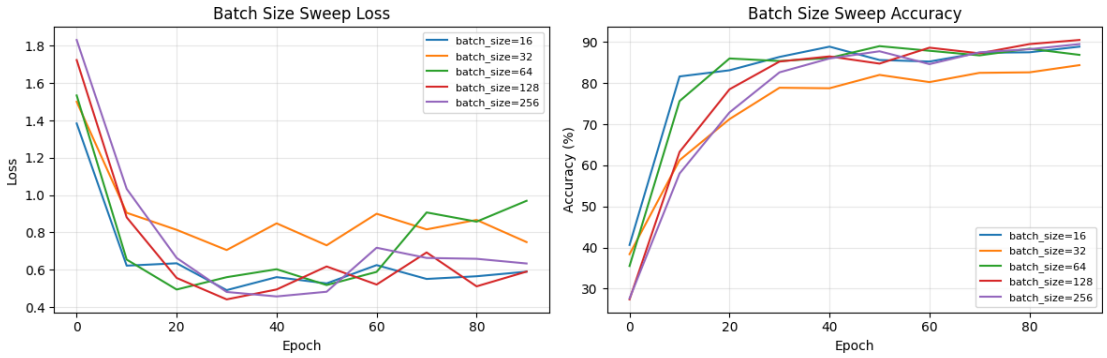
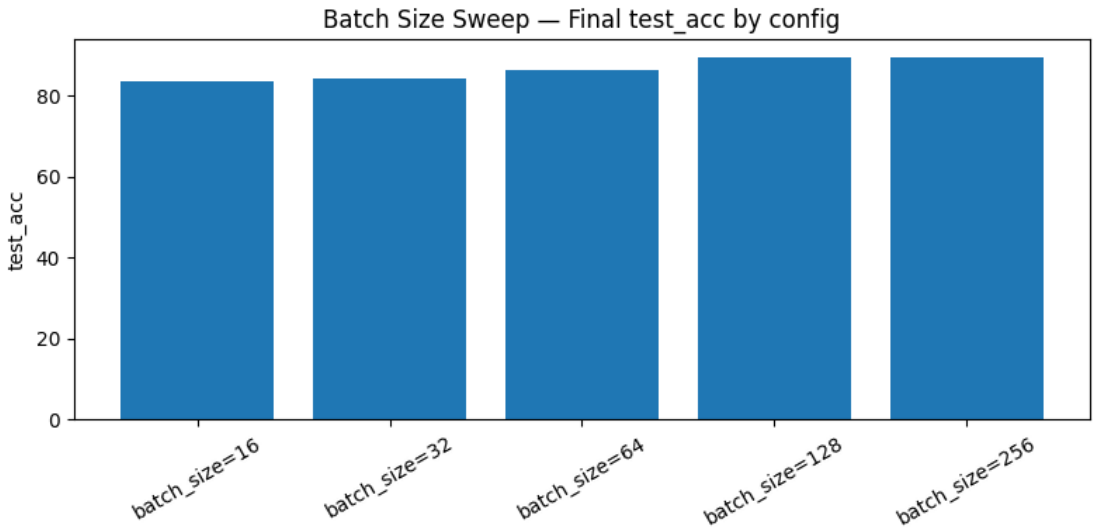
```
val_acc 63.2
epoch 20, dt 1.00s, train_loss 5.416e-01, val_loss 5.562e-01, train_acc 86.7,
val_acc 78.5
epoch 30, dt 1.34s, train_loss 2.043e-01, val_loss 4.414e-01, train_acc 94.4,
val_acc 85.2
epoch 40, dt 1.41s, train_loss 1.295e-01, val_loss 4.946e-01, train_acc 97.7,
val_acc 86.5
epoch 50, dt 1.17s, train_loss 7.512e-02, val_loss 6.178e-01, train_acc 95.4,
val_acc 84.8
epoch 60, dt 1.06s, train_loss 4.163e-02, val_loss 5.209e-01, train_acc 98.9,
val_acc 88.6
epoch 70, dt 1.11s, train_loss 8.539e-02, val_loss 6.929e-01, train_acc 97.2,
val_acc 87.2
epoch 80, dt 1.06s, train_loss 4.339e-03, val_loss 5.112e-01, train_acc 99.9,
val_acc 89.5
epoch 90, dt 1.21s, train_loss 1.164e-03, val_loss 5.903e-01, train_acc 100.0,
val_acc 90.5
Final result: train_loss 5.079e-04, val_loss 5.903e-01, test_loss 6.815e-01,
train_acc 100.0, val_acc 90.5, test_acc 89.5

--- [batch_size] Config: batch_size=256 ---
epoch 0, dt 0.09s, train_loss 1.889e+00, val_loss 1.831e+00, train_acc 28.2,
val_acc 27.6
epoch 10, dt 1.38s, train_loss 7.671e-01, val_loss 1.033e+00, train_acc 62.3,
val_acc 58.0
epoch 20, dt 0.86s, train_loss 6.058e-01, val_loss 6.634e-01, train_acc 80.7,
val_acc 72.9
epoch 30, dt 0.78s, train_loss 1.692e-01, val_loss 4.812e-01, train_acc 91.4,
val_acc 82.6
epoch 40, dt 0.87s, train_loss 1.389e-01, val_loss 4.573e-01, train_acc 96.0,
val_acc 86.0
epoch 50, dt 1.04s, train_loss 7.741e-02, val_loss 4.824e-01, train_acc 98.6,
val_acc 87.8
epoch 60, dt 0.74s, train_loss 4.020e-02, val_loss 7.181e-01, train_acc 95.3,
val_acc 84.6
epoch 70, dt 0.88s, train_loss 2.184e-02, val_loss 6.634e-01, train_acc 99.3,
val_acc 87.5
epoch 80, dt 0.87s, train_loss 1.609e-02, val_loss 6.589e-01, train_acc 99.1,
val_acc 88.2
epoch 90, dt 1.08s, train_loss 2.152e-03, val_loss 6.338e-01, train_acc 100.0,
val_acc 89.5
Final result: train_loss 1.375e-03, val_loss 6.338e-01, test_loss 6.713e-01,
train_acc 100.0, val_acc 89.5, test_acc 89.4
```






```
--- [epochs] Config: epochs=10 ---
epoch 0, dt 0.21s, train_loss 1.672e+00, val_loss 1.642e+00, train_acc 36.2,
val_acc 36.4
Final result: train_loss 5.933e-01, val_loss 1.642e+00, test_loss 6.340e-01,
train_acc 36.2, val_acc 36.4, test_acc 73.8

--- [epochs] Config: epochs=50 ---
epoch 0, dt 0.19s, train_loss 1.440e+00, val_loss 1.484e+00, train_acc 41.5,
val_acc 37.8
epoch 10, dt 1.66s, train_loss 7.613e-01, val_loss 8.816e-01, train_acc 69.1,
val_acc 63.6
epoch 20, dt 1.50s, train_loss 4.010e-01, val_loss 7.174e-01, train_acc 87.8,
val_acc 76.6
epoch 30, dt 1.51s, train_loss 1.643e-01, val_loss 4.997e-01, train_acc 95.7,
```

```
val_acc 86.9
epoch 40, dt 1.55s, train_loss 1.461e-01, val_loss 6.993e-01, train_acc 95.8,
val_acc 86.4
Final result: train_loss 1.873e-02, val_loss 6.993e-01, test_loss 4.952e-01,
train_acc 95.8, val_acc 86.4, test_acc 88.4

--- [epochs] Config: epochs=100 ---
epoch 0, dt 0.14s, train_loss 1.503e+00, val_loss 1.484e+00, train_acc 40.5,
val_acc 40.1
epoch 10, dt 1.31s, train_loss 7.558e-01, val_loss 8.053e-01, train_acc 70.1,
val_acc 64.5
epoch 20, dt 1.56s, train_loss 2.737e-01, val_loss 5.700e-01, train_acc 88.7,
val_acc 79.5
epoch 30, dt 3.26s, train_loss 1.648e-01, val_loss 4.857e-01, train_acc 94.6,
val_acc 87.0
epoch 40, dt 2.67s, train_loss 6.230e-02, val_loss 4.466e-01, train_acc 97.0,
val_acc 88.0
epoch 50, dt 3.27s, train_loss 1.617e-01, val_loss 5.594e-01, train_acc 96.8,
val_acc 86.1
epoch 60, dt 2.94s, train_loss 1.284e-01, val_loss 5.990e-01, train_acc 97.1,
val_acc 87.2
epoch 70, dt 2.03s, train_loss 8.027e-02, val_loss 6.296e-01, train_acc 97.5,
val_acc 86.8
epoch 80, dt 2.00s, train_loss 1.135e-02, val_loss 5.396e-01, train_acc 100.0,
val_acc 90.2
epoch 90, dt 2.14s, train_loss 9.694e-04, val_loss 6.246e-01, train_acc 100.0,
val_acc 89.9
Final result: train_loss 1.446e-04, val_loss 6.246e-01, test_loss 9.174e-01,
train_acc 100.0, val_acc 89.9, test_acc 88.3

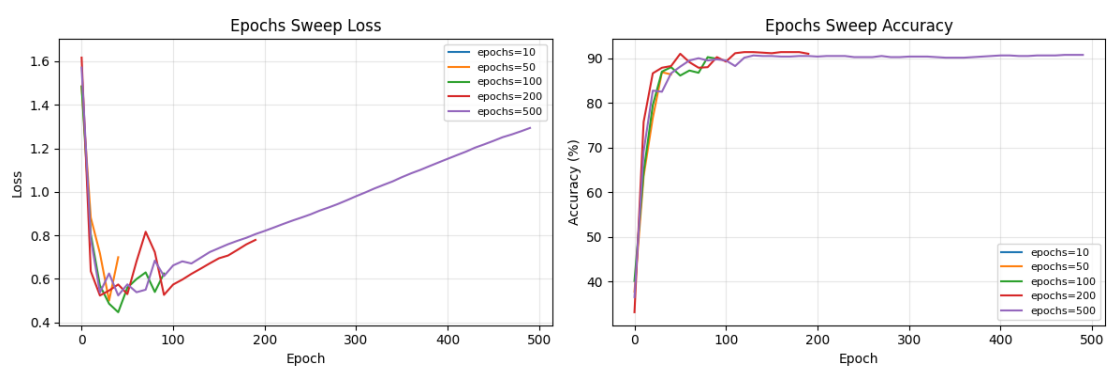
--- [epochs] Config: epochs=200 ---
epoch 0, dt 0.23s, train_loss 1.641e+00, val_loss 1.617e+00, train_acc 34.7,
val_acc 33.1
epoch 10, dt 2.19s, train_loss 3.295e-01, val_loss 6.351e-01, train_acc 85.5,
val_acc 75.8
epoch 20, dt 2.16s, train_loss 7.910e-02, val_loss 5.234e-01, train_acc 95.1,
val_acc 86.6
epoch 30, dt 2.12s, train_loss 1.004e-01, val_loss 5.463e-01, train_acc 97.2,
val_acc 87.9
epoch 40, dt 2.03s, train_loss 2.034e-01, val_loss 5.742e-01, train_acc 97.4,
val_acc 88.2
epoch 50, dt 1.87s, train_loss 9.154e-02, val_loss 5.298e-01, train_acc 99.4,
val_acc 91.0
epoch 60, dt 2.09s, train_loss 2.123e-01, val_loss 6.804e-01, train_acc 98.1,
val_acc 89.1
epoch 70, dt 2.26s, train_loss 3.209e-02, val_loss 8.163e-01, train_acc 97.0,
val_acc 87.9
epoch 80, dt 2.36s, train_loss 4.970e-02, val_loss 7.228e-01, train_acc 97.3,
```

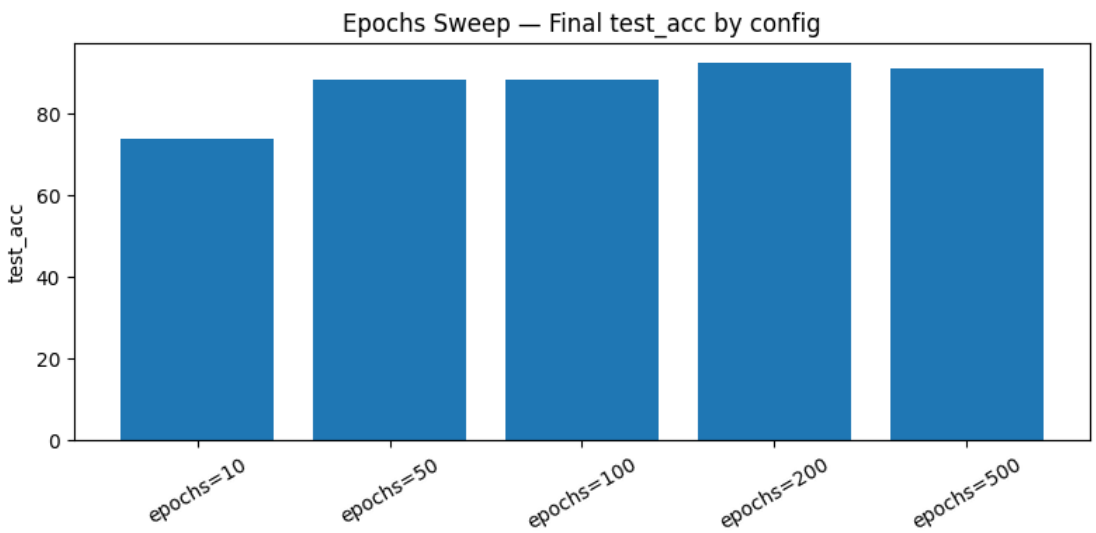
```
val_acc 88.0
epoch 90, dt 3.96s, train_loss 1.804e-03, val_loss 5.263e-01, train_acc 98.9,
val_acc 90.2
epoch 100, dt 2.56s, train_loss 9.787e-03, val_loss 5.739e-01, train_acc 99.2,
val_acc 89.2
epoch 110, dt 2.19s, train_loss 7.045e-05, val_loss 5.965e-01, train_acc 100.0,
val_acc 91.1
epoch 120, dt 3.12s, train_loss 4.051e-05, val_loss 6.229e-01, train_acc 100.0,
val_acc 91.4
epoch 130, dt 3.06s, train_loss 8.165e-05, val_loss 6.466e-01, train_acc 100.0,
val_acc 91.4
epoch 140, dt 3.30s, train_loss 1.565e-04, val_loss 6.711e-01, train_acc 100.0,
val_acc 91.2
epoch 150, dt 3.56s, train_loss 6.017e-05, val_loss 6.944e-01, train_acc 100.0,
val_acc 91.1
epoch 160, dt 2.13s, train_loss 2.016e-05, val_loss 7.071e-01, train_acc 100.0,
val_acc 91.4
epoch 170, dt 2.37s, train_loss 2.419e-05, val_loss 7.323e-01, train_acc 100.0,
val_acc 91.4
epoch 180, dt 2.60s, train_loss 9.925e-06, val_loss 7.585e-01, train_acc 100.0,
val_acc 91.4
epoch 190, dt 2.88s, train_loss 7.108e-06, val_loss 7.790e-01, train_acc 100.0,
val_acc 91.0
Final result: train_loss 1.271e-05, val_loss 7.790e-01, test_loss 7.054e-01,
train_acc 100.0, val_acc 91.0, test_acc 92.5

--- [epochs] Config: epochs=500 ---
epoch 0, dt 0.27s, train_loss 1.539e+00, val_loss 1.571e+00, train_acc 37.8,
val_acc 36.5
epoch 10, dt 2.63s, train_loss 7.504e-01, val_loss 7.844e-01, train_acc 74.6,
val_acc 69.1
epoch 20, dt 2.51s, train_loss 3.953e-01, val_loss 5.370e-01, train_acc 90.5,
val_acc 82.8
epoch 30, dt 1.83s, train_loss 1.435e-01, val_loss 6.248e-01, train_acc 90.9,
val_acc 82.5
epoch 40, dt 1.72s, train_loss 1.709e-01, val_loss 5.238e-01, train_acc 96.2,
val_acc 86.6
epoch 50, dt 1.53s, train_loss 7.915e-02, val_loss 5.748e-01, train_acc 95.6,
val_acc 88.1
epoch 60, dt 1.67s, train_loss 1.883e-02, val_loss 5.385e-01, train_acc 99.9,
val_acc 89.5
epoch 70, dt 1.47s, train_loss 4.516e-02, val_loss 5.502e-01, train_acc 98.2,
val_acc 90.0
epoch 80, dt 1.41s, train_loss 8.452e-03, val_loss 6.840e-01, train_acc 98.5,
val_acc 89.5
epoch 90, dt 1.42s, train_loss 7.853e-02, val_loss 6.126e-01, train_acc 99.7,
val_acc 89.8
epoch 100, dt 1.53s, train_loss 1.473e-02, val_loss 6.617e-01, train_acc 99.0,
```

```
val_acc 89.5
epoch 110, dt 1.67s, train_loss 8.497e-02, val_loss 6.805e-01, train_acc 98.5,
val_acc 88.2
epoch 120, dt 1.43s, train_loss 9.825e-04, val_loss 6.707e-01, train_acc 100.0,
val_acc 90.1
epoch 130, dt 1.56s, train_loss 3.266e-04, val_loss 6.968e-01, train_acc 100.0,
val_acc 90.6
epoch 140, dt 1.41s, train_loss 2.312e-04, val_loss 7.231e-01, train_acc 100.0,
val_acc 90.5
epoch 150, dt 1.43s, train_loss 9.336e-05, val_loss 7.411e-01, train_acc 100.0,
val_acc 90.5
epoch 160, dt 1.62s, train_loss 5.728e-05, val_loss 7.591e-01, train_acc 100.0,
val_acc 90.4
epoch 170, dt 1.35s, train_loss 1.751e-05, val_loss 7.744e-01, train_acc 100.0,
val_acc 90.4
epoch 180, dt 1.67s, train_loss 2.471e-05, val_loss 7.887e-01, train_acc 100.0,
val_acc 90.5
epoch 190, dt 1.41s, train_loss 2.896e-05, val_loss 8.056e-01, train_acc 100.0,
val_acc 90.5
epoch 200, dt 1.31s, train_loss 8.723e-06, val_loss 8.202e-01, train_acc 100.0,
val_acc 90.4
epoch 210, dt 1.36s, train_loss 3.870e-05, val_loss 8.358e-01, train_acc 100.0,
val_acc 90.5
epoch 220, dt 1.59s, train_loss 4.640e-06, val_loss 8.518e-01, train_acc 100.0,
val_acc 90.5
epoch 230, dt 1.48s, train_loss 1.017e-05, val_loss 8.672e-01, train_acc 100.0,
val_acc 90.5
epoch 240, dt 1.54s, train_loss 4.424e-06, val_loss 8.814e-01, train_acc 100.0,
val_acc 90.2
epoch 250, dt 1.60s, train_loss 7.819e-06, val_loss 8.961e-01, train_acc 100.0,
val_acc 90.2
epoch 260, dt 1.49s, train_loss 4.558e-06, val_loss 9.134e-01, train_acc 100.0,
val_acc 90.2
epoch 270, dt 1.78s, train_loss 6.076e-06, val_loss 9.282e-01, train_acc 100.0,
val_acc 90.5
epoch 280, dt 1.53s, train_loss 1.788e-06, val_loss 9.440e-01, train_acc 100.0,
val_acc 90.2
epoch 290, dt 1.53s, train_loss 8.959e-07, val_loss 9.615e-01, train_acc 100.0,
val_acc 90.2
epoch 300, dt 1.59s, train_loss 1.352e-06, val_loss 9.799e-01, train_acc 100.0,
val_acc 90.4
epoch 310, dt 1.70s, train_loss 5.979e-07, val_loss 9.972e-01, train_acc 100.0,
val_acc 90.4
epoch 320, dt 1.84s, train_loss 6.482e-07, val_loss 1.015e+00, train_acc 100.0,
val_acc 90.4
epoch 330, dt 1.57s, train_loss 1.088e-06, val_loss 1.032e+00, train_acc 100.0,
val_acc 90.2
epoch 340, dt 1.50s, train_loss 1.583e-07, val_loss 1.048e+00, train_acc 100.0,
```

```
val_acc 90.1
epoch 350, dt 1.50s, train_loss 3.017e-07, val_loss 1.067e+00, train_acc 100.0,
val_acc 90.1
epoch 360, dt 1.51s, train_loss 2.962e-07, val_loss 1.085e+00, train_acc 100.0,
val_acc 90.1
epoch 370, dt 1.91s, train_loss 2.570e-07, val_loss 1.101e+00, train_acc 100.0,
val_acc 90.2
epoch 380, dt 2.14s, train_loss 2.570e-07, val_loss 1.118e+00, train_acc 100.0,
val_acc 90.4
epoch 390, dt 2.34s, train_loss 1.192e-07, val_loss 1.135e+00, train_acc 100.0,
val_acc 90.5
epoch 400, dt 1.83s, train_loss 1.024e-07, val_loss 1.152e+00, train_acc 100.0,
val_acc 90.6
epoch 410, dt 1.62s, train_loss 1.360e-07, val_loss 1.168e+00, train_acc 100.0,
val_acc 90.6
epoch 420, dt 1.78s, train_loss 4.098e-08, val_loss 1.185e+00, train_acc 100.0,
val_acc 90.5
epoch 430, dt 2.07s, train_loss 4.284e-08, val_loss 1.203e+00, train_acc 100.0,
val_acc 90.5
epoch 440, dt 1.86s, train_loss 1.490e-08, val_loss 1.218e+00, train_acc 100.0,
val_acc 90.6
epoch 450, dt 1.85s, train_loss 2.421e-08, val_loss 1.234e+00, train_acc 100.0,
val_acc 90.6
epoch 460, dt 2.22s, train_loss 1.490e-08, val_loss 1.251e+00, train_acc 100.0,
val_acc 90.6
epoch 470, dt 2.06s, train_loss 1.118e-08, val_loss 1.264e+00, train_acc 100.0,
val_acc 90.8
epoch 480, dt 2.01s, train_loss 5.588e-09, val_loss 1.278e+00, train_acc 100.0,
val_acc 90.8
epoch 490, dt 1.89s, train_loss 9.313e-09, val_loss 1.294e+00, train_acc 100.0,
val_acc 90.8
Final result: train_loss 9.313e-09, val_loss 1.294e+00, test_loss 1.276e+00,
train_acc 100.0, val_acc 90.8, test_acc 90.9
```





[4]:
```
def run_sweep_class_hyperparameter(data, model_fn, configs: list[dict],␣
↪sweep_name: str = "") -> dict[str, dict]:
    all_results = {}
    for cfg in configs:
        label = ", ".join(f"{k}={v}" for k, v in cfg.items())
        print(f"\n--- [{sweep_name}] Config: {label} ---")
        model = model_fn(**cfg)          # pass config to constructor
        results = train_model(data, model)  # no **cfg here
        results["num_params"] = count_model_params(model)
        all_results[label] = results
    return all_results

def compute_linear_in(input_size=40, channels=25, kernel_size=3, stride=2,␣
↪padding=1):
    size = input_size
    # block 1: 2 convs + pool
    for _ in range(2):
        size = (size + 2*padding - kernel_size) // stride + 1
    size = size // 2
    # block 2: 2 convs + pool
    for _ in range(2):
        size = (size + 2*padding - kernel_size) // stride + 1
    size = size // 2
    return size * channels

#--- Kernel Size sweep ---
kernel_sizes = [1, 2, 3]
```

```
configs = [{"kernel_size": int(k), "linear_in":␣
↪compute_linear_in(kernel_size=int(k))} for k in kernel_sizes]
sweep_results["kernel_size"] = run_sweep_class_hyperparameter(data, SimpleCNN,␣
↪configs, sweep_name="kernel_size")
plot_sweep(sweep_results["kernel_size"], metric="val_acc", title_prefix="Kernel␣
↪Size Sweep")
plot_final_comparison(sweep_results["kernel_size"], metric="test_acc",␣
↪title_prefix="Kernel Size Sweep")

# -- No. Channels sweep ---
channel_list = [i for i in range(5, 25, 5)]
configs = [{"channels": int(c), "linear_in": int(c)} for c in channel_list]
sweep_results["channels"]    = run_sweep_class_hyperparameter(data, SimpleCNN,␣
↪configs, sweep_name="channels")
plot_sweep(sweep_results["channels"], metric="val_acc", title_prefix="Channels␣
↪Sweep")
plot_final_comparison(sweep_results["channels"], metric="test_acc",␣
↪title_prefix="Channels Sweep")

# -- Stride sweep ---
strides = [1, 2]
configs = [{"stride": int(s), "linear_in": compute_linear_in(stride=int(s))}␣
↪for s in strides]
sweep_results["stride"] = run_sweep_class_hyperparameter(data, SimpleCNN,␣
↪configs, sweep_name="stride")
plot_sweep(sweep_results["stride"], metric="val_acc", title_prefix="Stride␣
↪Sweep")
plot_final_comparison(sweep_results["stride"], metric="test_acc",␣
↪title_prefix="Stride Sweep")
```

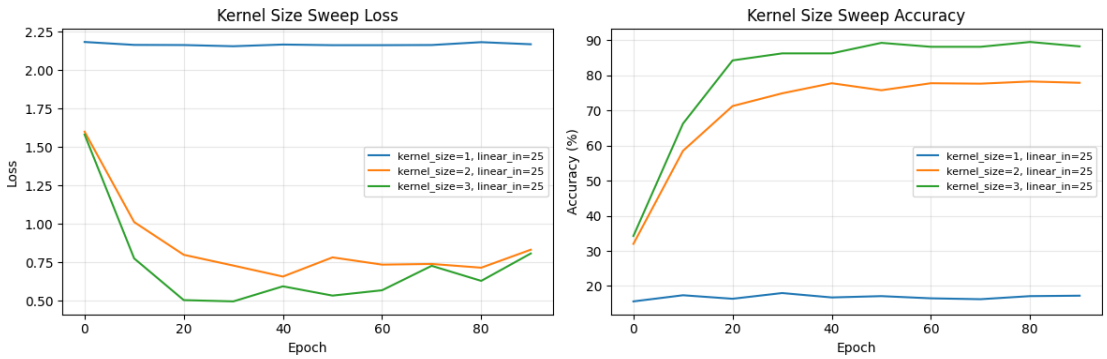
```
--- [kernel_size] Config: kernel_size=1, linear_in=25 ---
epoch 0, dt 0.52s, train_loss 2.260e+00, val_loss 2.183e+00, train_acc 15.2,
val_acc 15.6
epoch 10, dt 2.16s, train_loss 2.216e+00, val_loss 2.164e+00, train_acc 16.2,
val_acc 17.4
epoch 20, dt 2.03s, train_loss 2.145e+00, val_loss 2.163e+00, train_acc 17.1,
val_acc 16.4
epoch 30, dt 2.43s, train_loss 2.200e+00, val_loss 2.156e+00, train_acc 17.2,
val_acc 18.0
epoch 40, dt 2.32s, train_loss 2.202e+00, val_loss 2.167e+00, train_acc 17.1,
val_acc 16.8
epoch 50, dt 2.50s, train_loss 2.096e+00, val_loss 2.162e+00, train_acc 17.7,
val_acc 17.1
epoch 60, dt 2.53s, train_loss 2.150e+00, val_loss 2.162e+00, train_acc 18.2,
val_acc 16.5
epoch 70, dt 2.45s, train_loss 2.135e+00, val_loss 2.163e+00, train_acc 17.7,
```

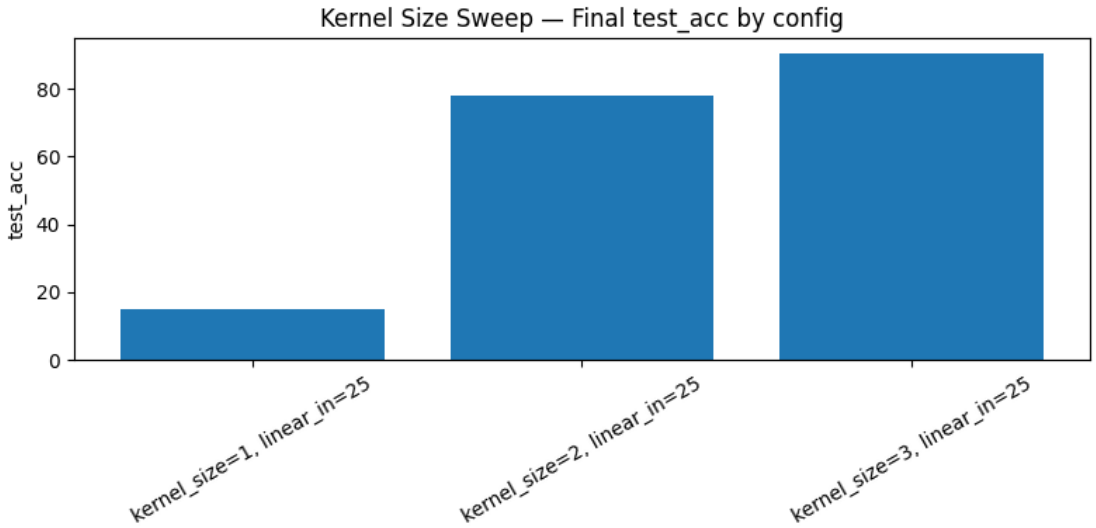
```
val_acc 16.2
epoch 80, dt 1.98s, train_loss 2.156e+00, val_loss 2.182e+00, train_acc 16.9,
val_acc 17.1
epoch 90, dt 1.68s, train_loss 2.161e+00, val_loss 2.168e+00, train_acc 17.3,
val_acc 17.2
Final result: train_loss 2.209e+00, val_loss 2.168e+00, test_loss 2.163e+00,
train_acc 17.3, val_acc 17.2, test_acc 14.8

--- [kernel_size] Config: kernel_size=2, linear_in=25 ---
epoch 0, dt 0.18s, train_loss 1.643e+00, val_loss 1.600e+00, train_acc 33.8,
val_acc 32.0
epoch 10, dt 1.78s, train_loss 8.353e-01, val_loss 1.012e+00, train_acc 64.0,
val_acc 58.5
epoch 20, dt 2.24s, train_loss 7.882e-01, val_loss 7.993e-01, train_acc 76.1,
val_acc 71.2
epoch 30, dt 1.80s, train_loss 5.453e-01, val_loss 7.292e-01, train_acc 81.5,
val_acc 74.9
epoch 40, dt 1.76s, train_loss 1.924e-01, val_loss 6.578e-01, train_acc 87.3,
val_acc 77.8
epoch 50, dt 1.66s, train_loss 3.787e-01, val_loss 7.824e-01, train_acc 85.8,
val_acc 75.8
epoch 60, dt 1.80s, train_loss 3.714e-01, val_loss 7.353e-01, train_acc 88.7,
val_acc 77.8
epoch 70, dt 1.62s, train_loss 3.314e-01, val_loss 7.397e-01, train_acc 88.5,
val_acc 77.6
epoch 80, dt 1.78s, train_loss 1.966e-01, val_loss 7.154e-01, train_acc 92.1,
val_acc 78.2
epoch 90, dt 1.87s, train_loss 2.760e-01, val_loss 8.324e-01, train_acc 90.7,
val_acc 77.9
Final result: train_loss 2.736e-01, val_loss 8.324e-01, test_loss 7.982e-01,
train_acc 90.7, val_acc 77.9, test_acc 78.1

--- [kernel_size] Config: kernel_size=3, linear_in=25 ---
epoch 0, dt 0.16s, train_loss 1.440e+00, val_loss 1.581e+00, train_acc 35.5,
val_acc 34.2
epoch 10, dt 2.27s, train_loss 5.232e-01, val_loss 7.756e-01, train_acc 74.2,
val_acc 66.2
epoch 20, dt 1.85s, train_loss 4.855e-01, val_loss 5.045e-01, train_acc 93.3,
val_acc 84.2
epoch 30, dt 1.71s, train_loss 1.076e-01, val_loss 4.962e-01, train_acc 95.8,
val_acc 86.2
epoch 40, dt 1.75s, train_loss 6.213e-02, val_loss 5.942e-01, train_acc 96.8,
val_acc 86.2
epoch 50, dt 1.63s, train_loss 6.734e-02, val_loss 5.339e-01, train_acc 99.1,
val_acc 89.2
epoch 60, dt 2.17s, train_loss 2.451e-02, val_loss 5.686e-01, train_acc 99.1,
val_acc 88.1
epoch 70, dt 2.05s, train_loss 6.348e-03, val_loss 7.276e-01, train_acc 99.0,
```

```
val_acc 88.1
epoch 80, dt 2.16s, train_loss 1.026e-02, val_loss 6.296e-01, train_acc 99.4,
val_acc 89.5
epoch 90, dt 1.65s, train_loss 3.699e-02, val_loss 8.079e-01, train_acc 97.7,
val_acc 88.2
Final result: train_loss 2.395e-02, val_loss 8.079e-01, test_loss 6.174e-01,
train_acc 97.7, val_acc 88.2, test_acc 90.3
```






```
--- [channels] Config: channels=5, linear_in=5 ---
epoch 0, dt 0.17s, train_loss 2.073e+00, val_loss 2.047e+00, train_acc 22.6,
val_acc 21.9
epoch 10, dt 1.63s, train_loss 1.117e+00, val_loss 1.279e+00, train_acc 49.6,
val_acc 46.5
epoch 20, dt 1.63s, train_loss 1.064e+00, val_loss 1.235e+00, train_acc 52.6,
val_acc 48.9
```

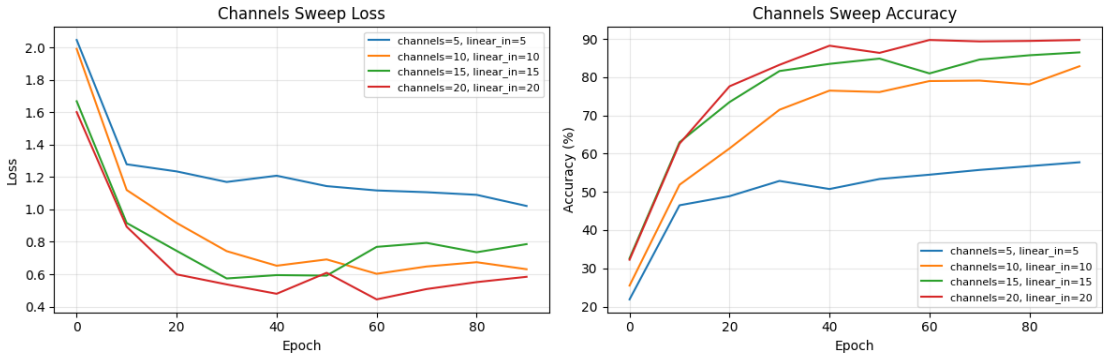
```
epoch 30, dt 1.64s, train_loss 1.097e+00, val_loss 1.170e+00, train_acc 54.1,
val_acc 52.9
epoch 40, dt 1.70s, train_loss 1.060e+00, val_loss 1.208e+00, train_acc 54.3,
val_acc 50.8
epoch 50, dt 1.88s, train_loss 9.528e-01, val_loss 1.144e+00, train_acc 55.9,
val_acc 53.4
epoch 60, dt 1.49s, train_loss 1.340e+00, val_loss 1.117e+00, train_acc 58.2,
val_acc 54.5
epoch 70, dt 1.46s, train_loss 1.196e+00, val_loss 1.106e+00, train_acc 59.5,
val_acc 55.8
epoch 80, dt 1.60s, train_loss 9.677e-01, val_loss 1.090e+00, train_acc 61.3,
val_acc 56.8
epoch 90, dt 1.51s, train_loss 9.100e-01, val_loss 1.021e+00, train_acc 64.1,
val_acc 57.8
Final result: train_loss 7.515e-01, val_loss 1.021e+00, test_loss 9.770e-01,
train_acc 64.1, val_acc 57.8, test_acc 60.8

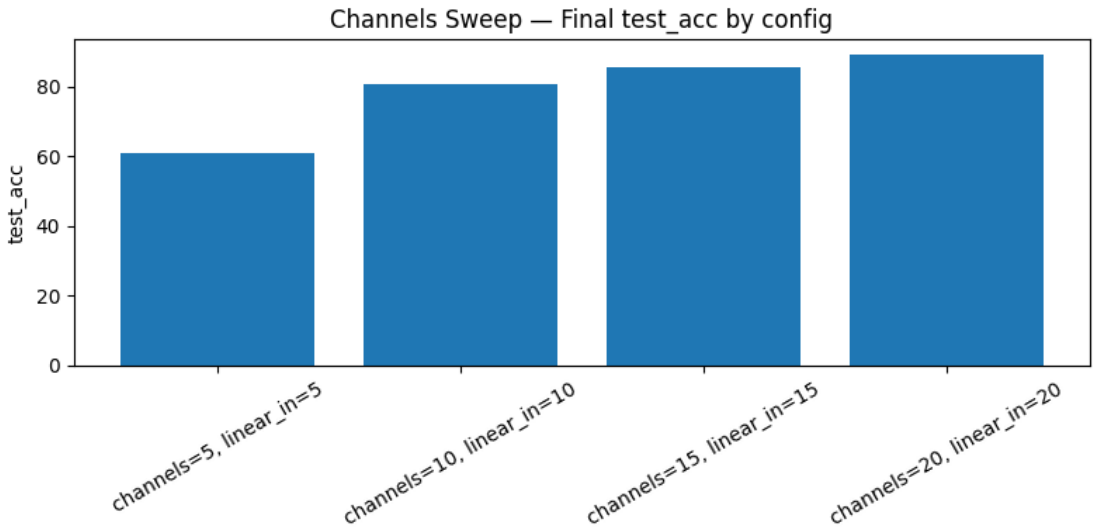
--- [channels] Config: channels=10, linear_in=10 ---
epoch 0, dt 0.15s, train_loss 1.975e+00, val_loss 1.992e+00, train_acc 26.2,
val_acc 25.5
epoch 10, dt 1.55s, train_loss 1.209e+00, val_loss 1.120e+00, train_acc 54.7,
val_acc 51.9
epoch 20, dt 1.47s, train_loss 8.295e-01, val_loss 9.164e-01, train_acc 66.5,
val_acc 61.4
epoch 30, dt 1.59s, train_loss 6.449e-01, val_loss 7.423e-01, train_acc 75.9,
val_acc 71.5
epoch 40, dt 1.65s, train_loss 4.379e-01, val_loss 6.523e-01, train_acc 82.0,
val_acc 76.5
epoch 50, dt 1.63s, train_loss 4.823e-01, val_loss 6.912e-01, train_acc 83.1,
val_acc 76.1
epoch 60, dt 1.43s, train_loss 4.387e-01, val_loss 6.025e-01, train_acc 86.2,
val_acc 79.0
epoch 70, dt 1.45s, train_loss 4.577e-01, val_loss 6.476e-01, train_acc 85.7,
val_acc 79.1
epoch 80, dt 1.58s, train_loss 4.972e-01, val_loss 6.740e-01, train_acc 85.9,
val_acc 78.1
epoch 90, dt 1.59s, train_loss 4.461e-01, val_loss 6.307e-01, train_acc 88.9,
val_acc 82.9
Final result: train_loss 2.037e-01, val_loss 6.307e-01, test_loss 6.995e-01,
train_acc 88.9, val_acc 82.9, test_acc 80.7

--- [channels] Config: channels=15, linear_in=15 ---
epoch 0, dt 0.15s, train_loss 1.752e+00, val_loss 1.668e+00, train_acc 34.1,
val_acc 32.6
epoch 10, dt 1.60s, train_loss 8.527e-01, val_loss 9.162e-01, train_acc 67.4,
val_acc 63.0
epoch 20, dt 1.46s, train_loss 4.824e-01, val_loss 7.441e-01, train_acc 82.1,
val_acc 73.5
```

```
epoch 30, dt 1.52s, train_loss 3.565e-01, val_loss 5.736e-01, train_acc 91.0,
val_acc 81.6
epoch 40, dt 1.54s, train_loss 1.499e-01, val_loss 5.944e-01, train_acc 94.8,
val_acc 83.5
epoch 50, dt 1.77s, train_loss 1.677e-01, val_loss 5.912e-01, train_acc 94.8,
val_acc 84.9
epoch 60, dt 1.49s, train_loss 1.434e-01, val_loss 7.688e-01, train_acc 92.5,
val_acc 81.0
epoch 70, dt 1.68s, train_loss 5.687e-02, val_loss 7.934e-01, train_acc 95.7,
val_acc 84.6
epoch 80, dt 1.39s, train_loss 6.059e-02, val_loss 7.355e-01, train_acc 97.1,
val_acc 85.8
epoch 90, dt 1.49s, train_loss 1.342e-01, val_loss 7.858e-01, train_acc 96.8,
val_acc 86.5
Final result: train_loss 1.439e-01, val_loss 7.858e-01, test_loss 8.177e-01,
train_acc 96.8, val_acc 86.5, test_acc 85.4

--- [channels] Config: channels=20, linear_in=20 ---
epoch 0, dt 0.19s, train_loss 1.603e+00, val_loss 1.601e+00, train_acc 33.1,
val_acc 32.2
epoch 10, dt 1.82s, train_loss 7.453e-01, val_loss 8.931e-01, train_acc 67.5,
val_acc 62.6
epoch 20, dt 1.63s, train_loss 3.048e-01, val_loss 5.988e-01, train_acc 85.9,
val_acc 77.6
epoch 30, dt 2.28s, train_loss 1.805e-01, val_loss 5.369e-01, train_acc 94.1,
val_acc 83.2
epoch 40, dt 1.60s, train_loss 9.076e-02, val_loss 4.791e-01, train_acc 98.1,
val_acc 88.2
epoch 50, dt 1.52s, train_loss 1.479e-01, val_loss 6.087e-01, train_acc 93.1,
val_acc 86.4
epoch 60, dt 1.70s, train_loss 5.985e-02, val_loss 4.443e-01, train_acc 99.8,
val_acc 89.8
epoch 70, dt 1.89s, train_loss 1.676e-03, val_loss 5.083e-01, train_acc 100.0,
val_acc 89.4
epoch 80, dt 2.51s, train_loss 9.472e-04, val_loss 5.511e-01, train_acc 100.0,
val_acc 89.5
epoch 90, dt 1.84s, train_loss 3.914e-04, val_loss 5.839e-01, train_acc 100.0,
val_acc 89.8
Final result: train_loss 2.014e-04, val_loss 5.839e-01, test_loss 7.474e-01,
train_acc 100.0, val_acc 89.8, test_acc 89.0
```

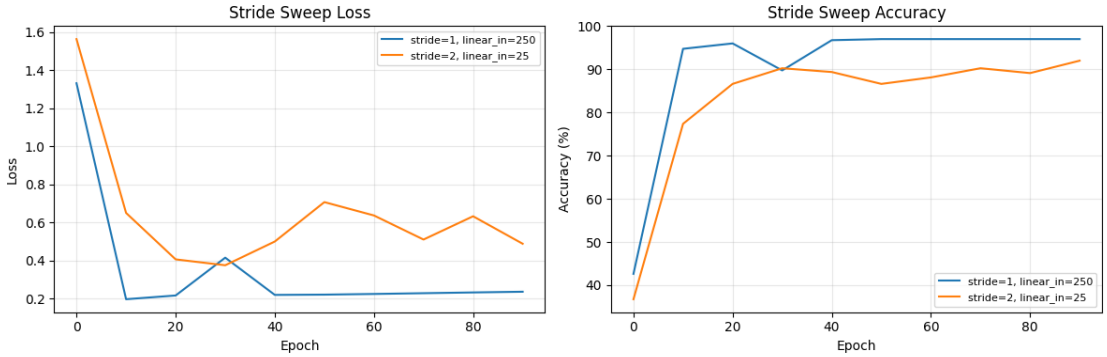


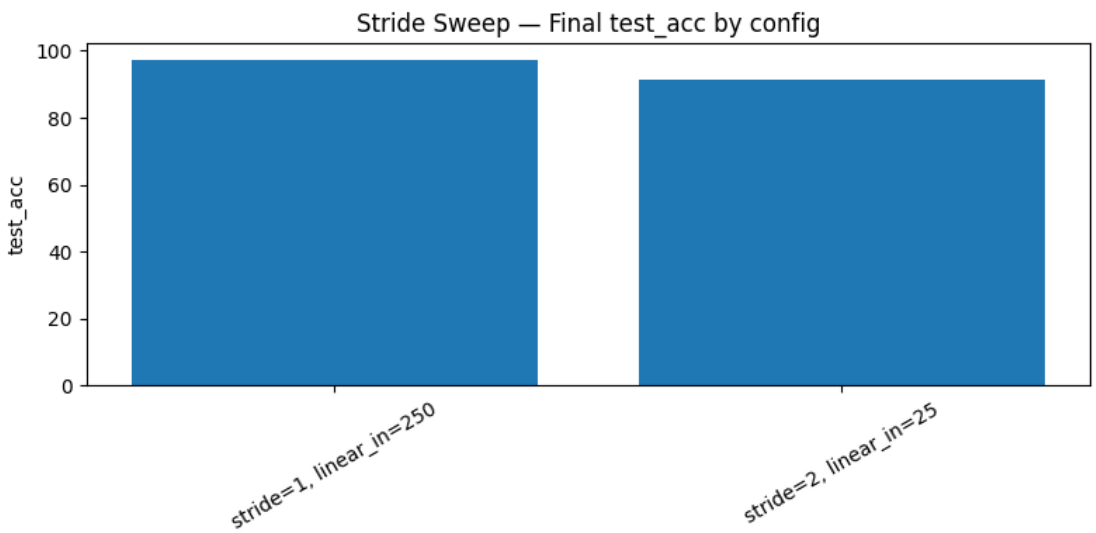



```
--- [stride] Config: stride=1, linear_in=250 ---
epoch 0, dt 0.37s, train_loss 1.344e+00, val_loss 1.331e+00, train_acc 44.0,
val_acc 42.6
epoch 10, dt 2.65s, train_loss 3.460e-02, val_loss 1.967e-01, train_acc 98.7,
val_acc 94.8
epoch 20, dt 2.32s, train_loss 7.154e-04, val_loss 2.162e-01, train_acc 99.3,
val_acc 96.0
epoch 30, dt 2.36s, train_loss 1.081e-01, val_loss 4.148e-01, train_acc 94.8,
val_acc 89.8
epoch 40, dt 2.16s, train_loss 4.139e-05, val_loss 2.193e-01, train_acc 100.0,
val_acc 96.8
epoch 50, dt 2.13s, train_loss 1.638e-05, val_loss 2.207e-01, train_acc 100.0,
val_acc 97.0
epoch 60, dt 2.39s, train_loss 6.436e-05, val_loss 2.243e-01, train_acc 100.0,
```

```
val_acc 97.0
epoch 70, dt 2.96s, train_loss 9.357e-06, val_loss 2.283e-01, train_acc 100.0,
val_acc 97.0
epoch 80, dt 2.53s, train_loss 2.107e-05, val_loss 2.323e-01, train_acc 100.0,
val_acc 97.0
epoch 90, dt 2.41s, train_loss 2.306e-06, val_loss 2.360e-01, train_acc 100.0,
val_acc 97.0
Final result: train_loss 1.098e-05, val_loss 2.360e-01, test_loss 2.030e-01,
train_acc 100.0, val_acc 97.0, test_acc 97.3

--- [stride] Config: stride=2, linear_in=25 ---
epoch 0, dt 0.16s, train_loss 1.653e+00, val_loss 1.562e+00, train_acc 39.2,
val_acc 36.8
epoch 10, dt 1.55s, train_loss 7.944e-01, val_loss 6.498e-01, train_acc 81.0,
val_acc 77.4
epoch 20, dt 1.68s, train_loss 2.816e-01, val_loss 4.059e-01, train_acc 93.0,
val_acc 86.6
epoch 30, dt 1.53s, train_loss 1.548e-01, val_loss 3.750e-01, train_acc 96.8,
val_acc 90.2
epoch 40, dt 1.61s, train_loss 1.529e-01, val_loss 4.996e-01, train_acc 98.1,
val_acc 89.4
epoch 50, dt 1.45s, train_loss 1.456e-01, val_loss 7.065e-01, train_acc 93.2,
val_acc 86.6
epoch 60, dt 1.46s, train_loss 5.451e-02, val_loss 6.364e-01, train_acc 97.7,
val_acc 88.1
epoch 70, dt 1.72s, train_loss 5.816e-03, val_loss 5.100e-01, train_acc 98.2,
val_acc 90.2
epoch 80, dt 1.83s, train_loss 3.791e-02, val_loss 6.322e-01, train_acc 98.5,
val_acc 89.1
epoch 90, dt 1.63s, train_loss 1.190e-03, val_loss 4.879e-01, train_acc 100.0,
val_acc 92.0
Final result: train_loss 3.575e-04, val_loss 4.879e-01, test_loss 6.271e-01,
train_acc 100.0, val_acc 92.0, test_acc 91.4
```





1.2 Summary Table

[5]:
```
summary_df = build_summary_table(sweep_results)
summary_df
```

[5]:
```
          sweep                       config  test_acc  val_acc_final
train_acc_final  val_loss_final  train_loss_final  num_params
0    batch_size               batch_size=128      89.5         90.500
100.00000        0.590306      5.078583e-04        6060
1    batch_size               batch_size=256      89.4         89.500
100.00000        0.633788      1.375056e-03        6060
2    batch_size                batch_size=64      86.5         86.875
97.03125        0.969574      1.913787e-02        6060
3    batch_size                batch_size=32      84.5         84.375
95.71875        0.748352      5.148472e-01        6060
4    batch_size                batch_size=16      83.6         88.875
97.96875        0.589902      1.388200e-01        6060
5      channels    channels=20, linear_in=20      89.0         89.750
100.00000        0.583869      2.014041e-04        3950
6      channels    channels=15, linear_in=15      85.4         86.500
96.81250        0.785753      1.439218e-01        2290
7      channels    channels=10, linear_in=10      80.7         82.875
88.90625        0.630663      2.037381e-01        1080
8      channels      channels=5, linear_in=5      60.8         57.750
64.09375        1.021388      7.515295e-01         320
9        epochs                    epochs=200      92.5         91.000
100.00000        0.778976      1.270722e-05        6060
10       epochs                    epochs=500      90.9         90.750
```

100.00000 1.293657 9.313225e-09 6060
11 epochs epochs=50 88.4 86.375
95.84375 0.699347 1.873023e-02 6060
12 epochs epochs=100 88.3 89.875
100.00000 0.624648 1.445593e-04 6060
13 epochs epochs=10 73.8 36.375
36.25000 1.642251 5.933174e-01 6060
14 kernel_size kernel_size=3, linear_in=25 90.3 88.250
97.68750 0.807877 2.395174e-02 6060
15 kernel_size kernel_size=2, linear_in=25 78.1 77.875
90.68750 0.832418 2.735904e-01 4160
16 kernel_size kernel_size=1, linear_in=25 14.8 17.250
17.34375 2.168446 2.209225e+00 2260
17 lr lr=0.01 88.8 86.625
97.59375 0.905998 9.662508e-02 6060
18 lr lr=0.00031622776601683794 58.4 55.375
64.96875 1.061899 7.111596e-01 6060
19 lr lr=9.999999999999999e-06 21.4 21.125
21.34375 1.991232 1.876058e+00 6060
20 lr lr=0.31622776601683794 10.6 10.000
9.93750 2.318876 2.338308e+00 6060
21 lr lr=10.0 9.8 9.875
9.84375 2.721964 2.589050e+00 6060
22 stride stride=1, linear_in=250 97.3 97.000
100.00000 0.236000 1.097705e-05 8310
23 stride stride=2, linear_in=25 91.4 92.000
100.00000 0.487919 3.575331e-04 6060